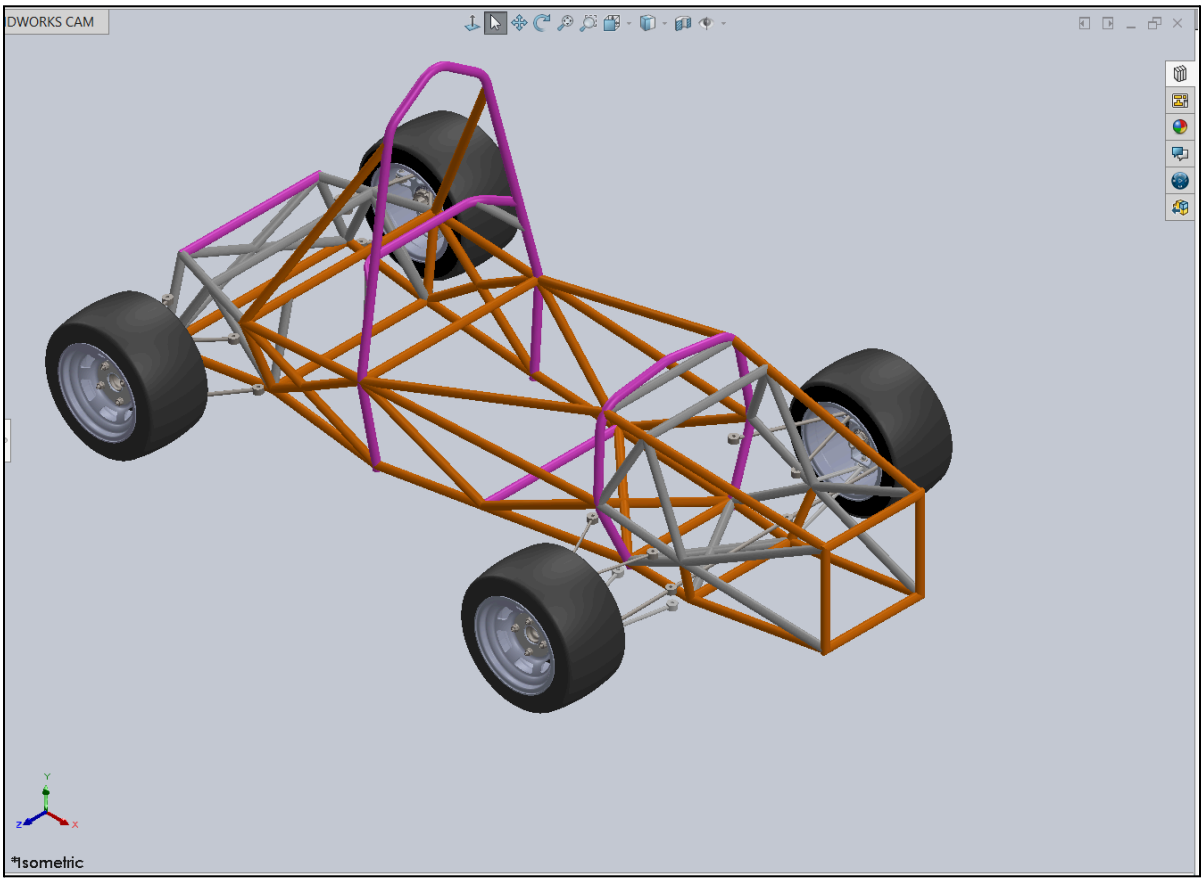
FSAE SUSPENSION DEVELOPMENT



Written by:
Elijah Akpan

Advisor:
Professor Michalson

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Abstract

After the successful 3 year run of EV24-EV26, WPI's FSAE team is building a new racecar that will be expected to match or exceed the performance of its predecessor. Vehicle dynamics, which deals with how a car handles and how effectively power can be put down, is one of the most important aspects of creating a vehicle. Without a proper suspension setup the vehicle will be uncontrollable, or in even worse cases unable to drive. The goal of this independent study project was to create the optimal suspension layout as well as design the outboard assembly (a-arms, push rods, and tie-rods) based on the pickup points from the suspension layout.

Suspension Layout

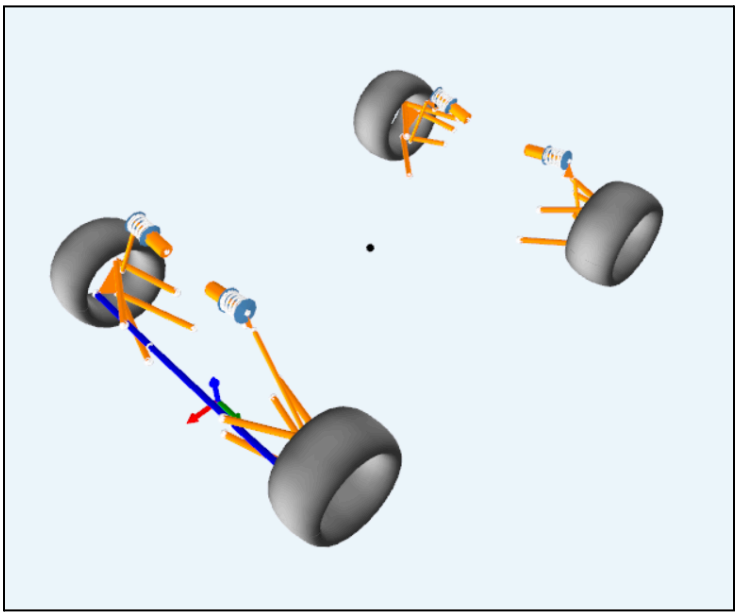


Figure 1: Suspension Layout from OptimumKinematics

Several goals were given for the suspension system before development. The first is that the suspension is rules compliant with both Formula Hybrid and FSAE Michigan. There are a couple of main rules stated by the FH+E and FSAE rulebook that every suspension system has to follow. The main rule that the suspension was designed around is V3.1.1 from FSAE Michigan rulebook which states “the vehicle must have a fully operational suspension system with shock absorbers, front and rear, with usable minimum wheel travel of 50 millimeters, with a driver seated.”

The suspension system was also designed around our tires. We currently run 16.0X7.5-10 Hoosier LC0 tires. Using TTC data we can get the direct behavior of the tires. As shown in figure 1, the LC0s lose lateral force at higher slip and camber angles. So another goal was to minimize camber gain as the racecar corners. In addition, the tires generate peak lateral force at equal slip angles as also shown by the graph below. This behavior led us to use parallel steering instead of exploring ackermann geometry.

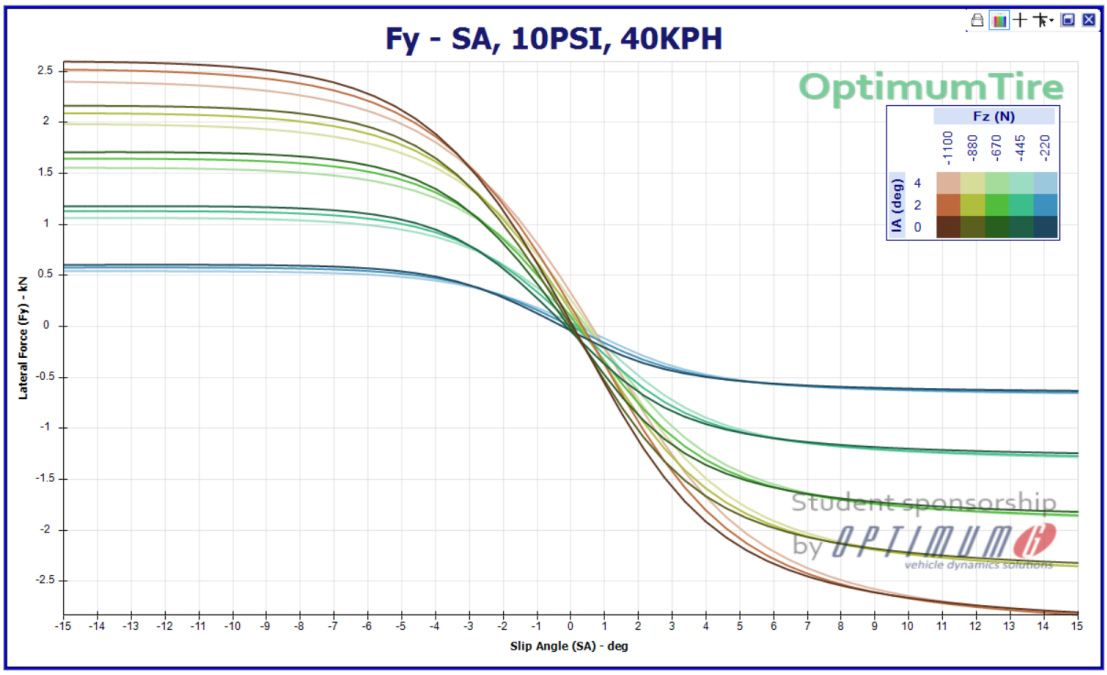


Figure 2: Lateral Force vs Slip Angle graph

Outboard Assembly

In addition to the rules, the design of EV27's suspension system was heavily influenced by EV24's push rod system, which led to many of our target values being based on it. We first started outboard, which are the pickup points that are furthest outwards and deal with the upright. These kinematics values are the caster angle, scrub radius, and caster trail. These values largely dictate how the vehicle is going to feel when driving, which is why we aimed for values close to EV24 so our driver has a similar feel when steering the racecar.

Scrub radius is the distance from the center of the contact patch to the point where the steering axis intersects the ground viewed from the front of the car. This is what determines the control that the driver has of the vehicle. The goal is to get as close to zero as possible but still have a slight positive scrub so the driver doesn't have to steer the

entire wheel. Therefore, a value of 6.15 millimeters (which is what our previous car had) was targeted. The final value that was achieved was 6.15 millimeters.

Mechanical trail, which is the distance between where the steering axis intersects the ground and the center of the contact patch viewed from the side of the car. A value of 9.807 millimeters was targeted for the mechanical trail (which is what our previous car had), and the final value achieved was 9.808 millimeters. Caster angle, which is the shifting of the steering axis from left or right from the center vertical line of the tire, was the next pickup point that was adjusted. A positive caster angle is preferred due to the steering axis being in front of the contact patch. This allows for steering to be in the direction that the car moves in. A value of 4.924 degrees was targeted for caster angle (which is what our previous car had), and the final value that was achieved was 4.925 degrees.

Roll Center

However, a very substantial difference between EV27 and EV24 is the size of the rear. Our previous car utilized a single motor, single differential setup. EV27 will have two motors and two gearboxes that attach to the rear axle of the car. This significantly increased the size and weight of the rear of the car. To adjust for this, we used OptimumKinematics to model our suspension with a wider rear. The wider rear caused our rear a-arms to significantly decrease in length to keep the track width at similar lengths. The shorter A-arms cause our roll center, which is the motion of the vehicle from side to side, to increase. The recommended roll gradient (which determines how much roll occurs per g of acceleration) for a high downforce racecar is between 0.2-0.7 deg/g. So in order for us to be within this range, several adjustments had to be made.

Firstly, a higher roll center that is closer to the center of gravity helps to induce less roll in corners. Less roll means less lateral load transfer which means the tires can effectively contact the ground more during

cornering. However, increasing the roll center height brings it further away from the ground. This will induce a jacking force onto the control arms which will cause the chassis to rise when cornering. Jacking force is a diagonal force from the contact patch that occurs when cornering in order to pull the car in the direction of the turn. This sideways force has a vertical component that increases as your roll center gets further away from the ground and closer to your roll center. To counter these jacking forces, an optimization was ran in OptimumKinematics, with the objective being set on roll center z heights and the boundary points being set on the a-arms as well.

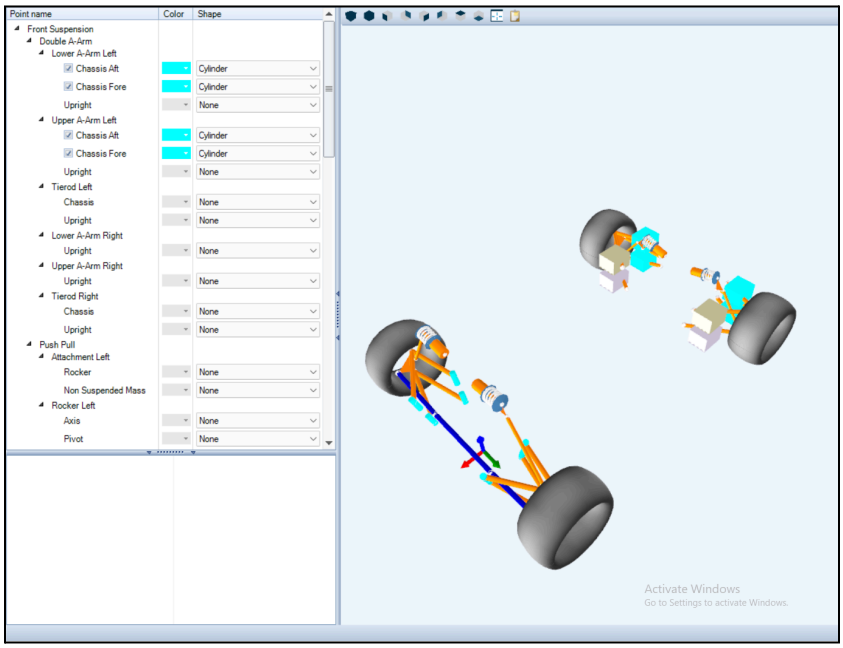


Figure 3: Optimization for Jacking forces

This optimization gave a front roll center of 60 millimeters above ground plane and a rear roll center center of 68 millimeters. Using a MATLAB script that calculates suspension parameters, this roll center gives a roll gradient of 0.738 deg/g which is close to the ideal range for a high downforce racecar.

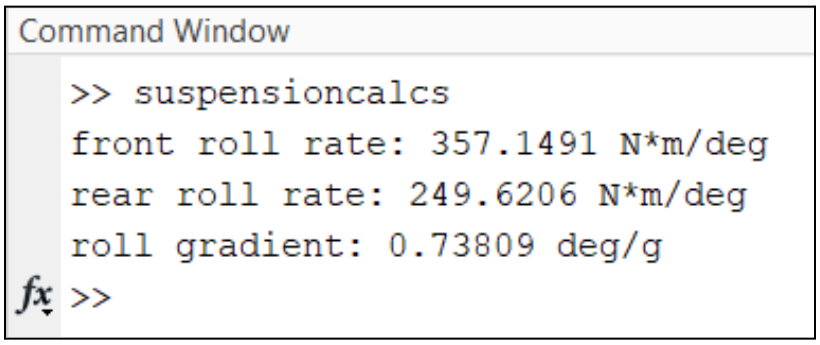


Figure 4: Roll gradient calculations from MATLAB

Front View Swing Arm

As stated previously, the LCOs lose lateral force as the camber and slip angle changes. Camber change rate is directly controlled by the front view swing arm (FVSA). The FVSA is the instant centre location of the steering axis intersecting with the center of the tire. This instantaneous center controls both roll center height and camber changes. So, another optimization had to be run to increase the front view swing axis to minimize camber change. The objective was set on 0 for camber gain in pitch with boundary points put on the a-arms. The resulting front view swing arm angle came out to 5.936 degrees for the front and 0.159 degrees for the rear. These angles give us a camber gain in pitch of -0.03 degrees, camber gain in steering of -0.018 degrees, camber angle gain in heave of -0.023 degrees, and camber angle gain in roll of -0.06.



Figure 5: Camber angle gain in pitch

Anti-geometry

The second difference was anti-geometry. Unlike our previous car, the suspension is now designed around our aerodynamics package. Anti-geometry, such as anti-dive, helps to reduce constant ride height changes when cornering and braking. It does this by having the suspension geometry direct forces into the chassis instead of just the spring/damper. This in turn allows the car to produce more downforce because air flow is stable and not constantly changing due to ride height staying consistent. Anti-squat is similar to anti-dive but now aids the rear in having constant ride height. With the new, more powerful, and heavier powertrain plus the wing, we needed to ensure the vehicle would not bottom out under hard acceleration. The layout currently has 39% anti-dive and 21% anti-squat. These numbers were derived from literature and hand calculations.

The higher anti-dive percentage was mainly because the front handles more force due to the high brake bias. Anti-geometry is created by changing the angle of the suspension a-arms (chassis pickup points)

from the side view of the car. Anti-geometry is measured as follows: draw an imaginary line from the upper and lower A-arms; where they intersect creates an instantaneous center. Then draw a line from the contact patch through the instantaneous center; measure where it intersects the center of gravity to get the anti-dive or anti-squat percentage. This intersection point has a vertical component that counters compression forces on the suspension. This is what helps maintain ride height.

Tie-rods/Bumpsteer

In addition to those differences, an optimization for near-zero bump and roll steer was done. Bump and roll steer can be detrimental to a suspension system due to both being toe gain in heave/roll. Toe change can cause unpredictable handling when the vehicle undergoes heave, pitch, and roll. In order to counter bump steer, the inner and outer tie rod have to be positioned so that they travel on the same arc as themselves and the A-arms. So for this, an optimization with 1 inch of heave, boundary boxes on the inner and outer tie rod, and an objective of 0 degrees of toe gain in heave was run. As shown in the figure below, the goal of near-zero bump steer was achieved.

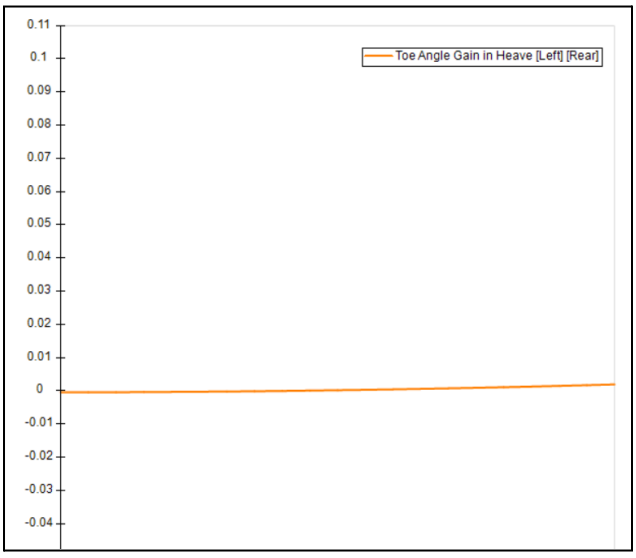


Figure 6: Toe angle gain in heave

Suspension Components

Uprights

After the suspension layout was completed (over 20 optimizations done), pickup points were imported into SolidWorks to model the suspension components and bring the layout to life. The first component that was designed using the pickup points was the uprights. The uprights connect the suspension components to the wheel. This component is mainly designed from the scrub radius, mechanical trail, caster angle, and tie rod pickup points. A distinct feature of the front uprights that differentiates them from the rear is the top pickup point. The top pickup point is closer to the chassis in the track width direction due to the 10 degrees of kingpin inclination that was included to get us to our target scrub radius. Many iterations of pickup point adjustments had to be done due to the 3D modeling exposing clearance issues. One adjustment was the tie rod needing to be brought slightly more inward due to previously only having 0.02 inches of clearance from the inner diameter of the rim.

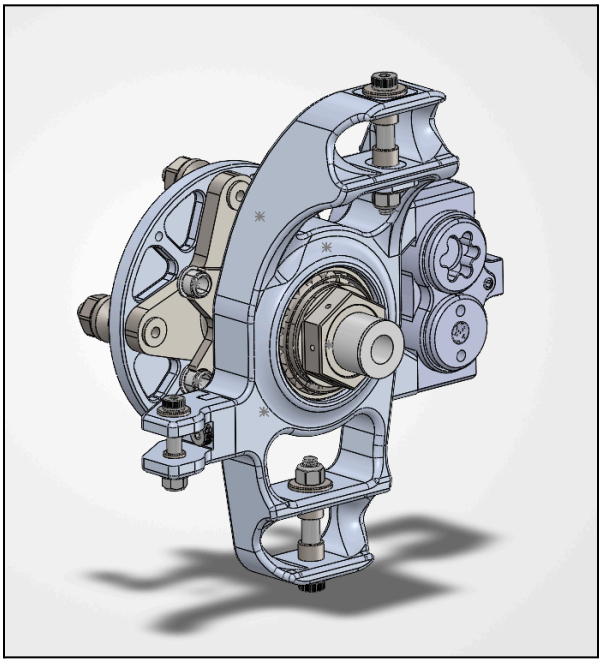


Figure 7: Front Upright Assembly

Control arms

Due to our new setup having anti-geometry, control arms will now see significantly more force than before. This emphasized the need for finite element analysis and simulation to get accurate force numbers from telemetry (IMU). The control arm shown below was created using hand calculations instead of telemetry. A force of 2654.82 newtons was calculated using longitudinal load transfer seen in a 2.1 g braking scenario (0.4(322*9.8*2.1)) and placed on the bearing cup attached to the upright.

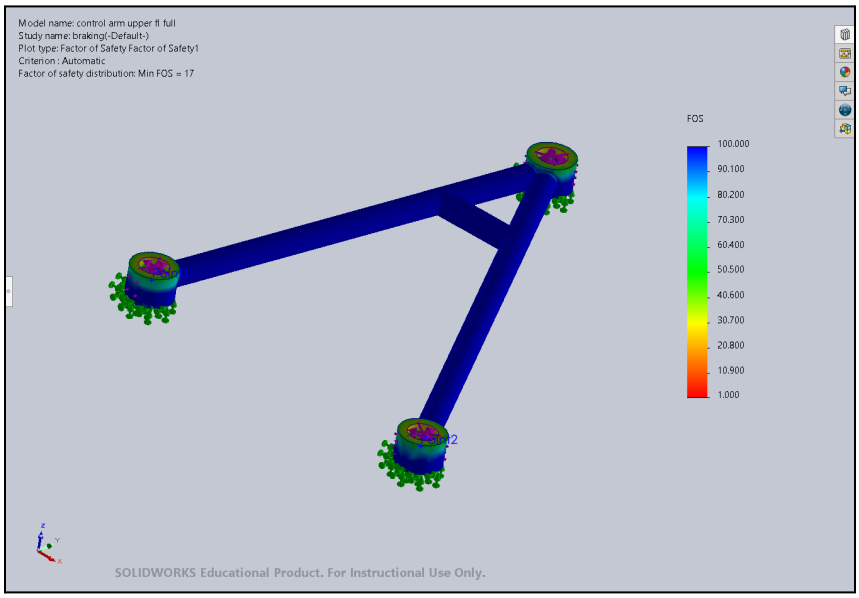


Figure 8: FEA of control arm

After a CDR with the team and alumni, the initial FEA of the control arms shown above had the wrong setup. This caused an artificially high factor of safety due to the wrong fixture being used. In addition, an alumnus mentioned that the bearing cup shape is largely dictated by the tabs themselves. So, the focus shifted to the design of the tabs before the bearing cup could be modified.

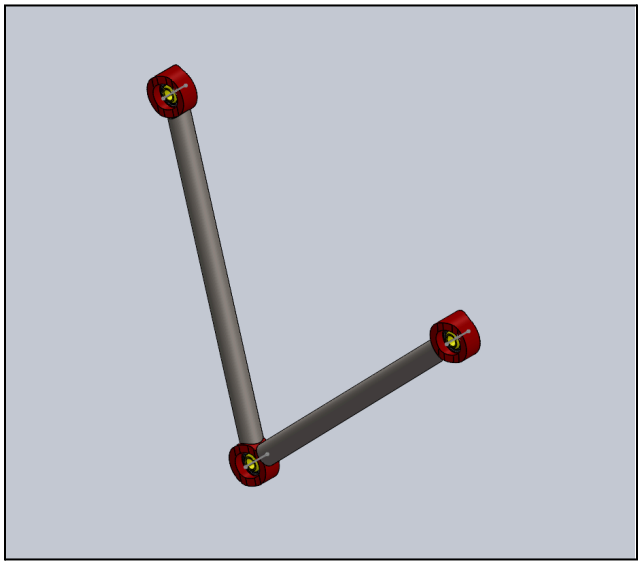


Figure 9: Updated Bearing Cup Design

As preliminary tab designs were created, the bearing cup design was changed to fit the tabs and provide proper clearance. Highlighted in red is the new bearing cup design that is almost identical to EV24's bearing cups. The tubing of the A-arm shown in Figure 7 has an outer diameter of 0.625 inches and a wall thickness of 0.083 inches. Further analysis needs to be done to perfect the design of the A-arms. An area of priority is to have an accurate finite element analysis setup. Also, gaining accurate numbers from telemetry data and simulation is another priority before further design adjustments can be made.

Interference Check

To ensure no collisions occur during the full suspension and steering travel, a mock rolling chassis with the base design of the suspension components had to be constructed. The base design of the suspension components was created using the pickup point measurements. A base design of the push-rods, tie-rods, steering rack, and control arms was put into an assembly with the chassis, then moved through their max travel ranges. The following figures showcase the components and the interference check.

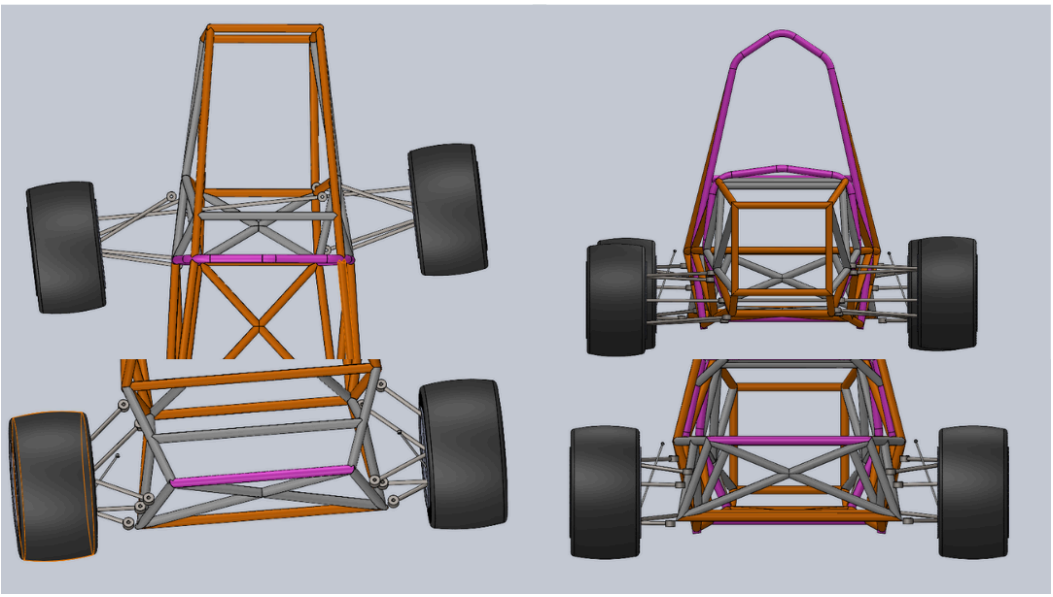


Figure 10: Interference check

Conclusion

The suspension layout is finished and contains values close to or at the target values. The outboard assembly of the suspension is still a work in progress, as more accurate simulation results will cause slight design changes, but the overall design is near completion. Tie rods and the push-rods are the next components to be designed after the control arms.